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| Table 14.3: Comparison of alcoholic fermentation and lactic acid fermentation | |
|---|---|
| **Alcoholic fermentation** | **Lactic acid fermentation** |
| 1. It produces alcohol and releases $CO_2$ from pyruvic acid. | It produces lactic acid and does not release $CO_2$ from pyruvic acid. |
| 2. It takes place in two steps. | It takes place in single step. |
| 3. It involves two enzymes, pyruvate decarboxylase with $Mg^{++}$ and alcohol dehydrogenase. | It uses one enzyme, lactate dehydrogenase with $Zn^{++}$. |
| 4. It forms acetaldehyde as intermediate compound. | Does not form any intermediate compound. |
| 5. It commonly occurs in yeast. | Occurs in bacteria, some fungi and vertebrate muscles. |

**Industrial uses of alcoholic fermentation:**

1. In bakeries, it is used for preparing bread, cakes, biscuits.
2. In beverage industries for preparing wine and alcoholic drinks.
3. In producing vinegar and in tanning, curing of leather.
4. Ethanol is used to make gasohol (a fuel that is used for cars in Brazil).

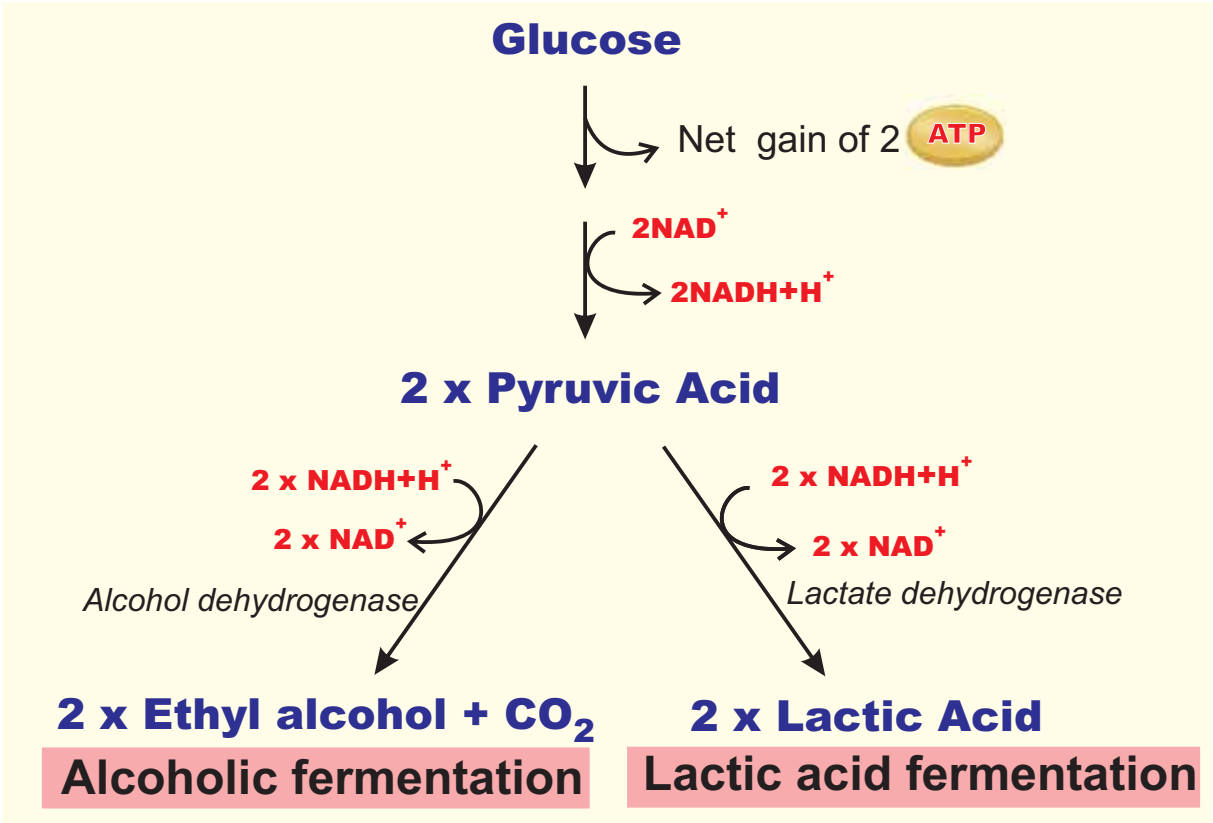



**Figure 14.12:** Anaerobic Respiration